

Synthetic Market Phase 18

Deterministic causal patching on the discovered market-only representation. This version compares four interventions on the same stratified 24-prompt cohort: the named one-dimensional `leader_axis` and `dispersion_axis`, plus the full top-4 `market_mean` subspace at L4 and L35, each against a matched orthogonal random control.

DATE	COHORT	CONTEXT	GENERATION
26 March 2026	24 prompts from 16 strata	market_only	temp=0.0, top_p=0.95, max_tokens=15000

MAIN READ

The 1D named-axis story is weak, but the 4D subspace story is stronger. Removing the named `leader_axis` changes asset choice in only 2/24 prompts, while its matched random control changes 7/24. Removing the named `dispersion_axis` ties its control on asset changes (3/24 each). But when the intervention expands to the full top-4 subspace, the targeted effect grows: L4 top-4 changes asset choice in 6/24 prompts versus 4/24 for its control, and L35 top-4 changes asset choice in 8/24 prompts versus 3/24 for its control, with one `buy_token` → `record_observation` flip. The market representation looks more like a distributed subspace than a single causal knob.

L4 1D ASSET CHANGE	L4 4D ASSET CHANGE	L35 1D ASSET CHANGE	L35 4D ASSET CHANGE
2 / 24	6 / 24	3 / 24	8 / 24
control = 7 / 24	control = 4 / 24	control = 3 / 24	control = 3 / 24

Methodology

All runs in this report use the same prompt cohort and generation surface:

- Context variant: `market_only`
- Selection strategy: stratified round-robin over (`family`, `family_variant`, `roster_key`)
- Sample size: 24 prompts from 16 strata

- Tool schema: trading_v1, tool choice: required

Parameter	Value
Temperature	0
Top-p	0.95
Top-k	-1
Max tokens	15000

The patch is applied to the market token span only:

1. locate the exact market span in the rendered chat prompt
2. compute the market_mean hidden state over that span
3. standardize with the Phase 15 residualized PCA mean and scale
4. remove either a named 1D component or the top 4 components at the target layer
5. convert that standardized delta back to raw hidden space
6. add the same raw delta to every token in the market span

The matched random control removes an equal-norm random subspace orthogonal to the targeted one.

Intervention	Layer	Dim	Target
Leader 1D	4	1D	leader_axis
Leader 4D	4	4D	top 4 market_mean PCs
Dispersion 1D	35	1D	dispersion_axis
Dispersion 4D	35	4D	top 4 market_mean PCs

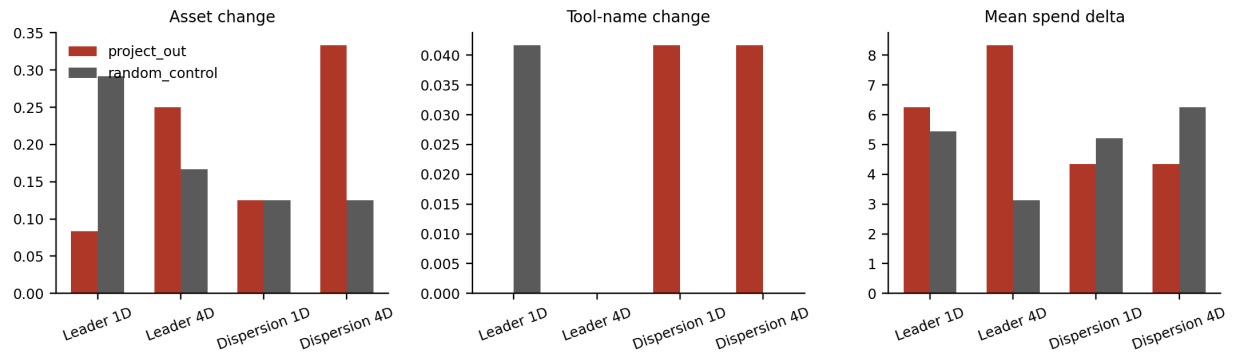
Patch Sanity

The mechanical side is clean across all four experiments.

Intervention	Project-out coeff after	Random-control coeff after	Rows with stats
Leader 1D @ L4	0.000005	11.706	24 / 24
Leader 4D @ L4	0.00001	47.532	24 / 24
Dispersion 1D @ L35	0.000001	0.804	24 / 24
Dispersion 4D @ L35	0.000003	9.055	24 / 24

The patch fires cleanly in all cases. The remaining question is whether removing each target perturbs behavior more than the matched control.

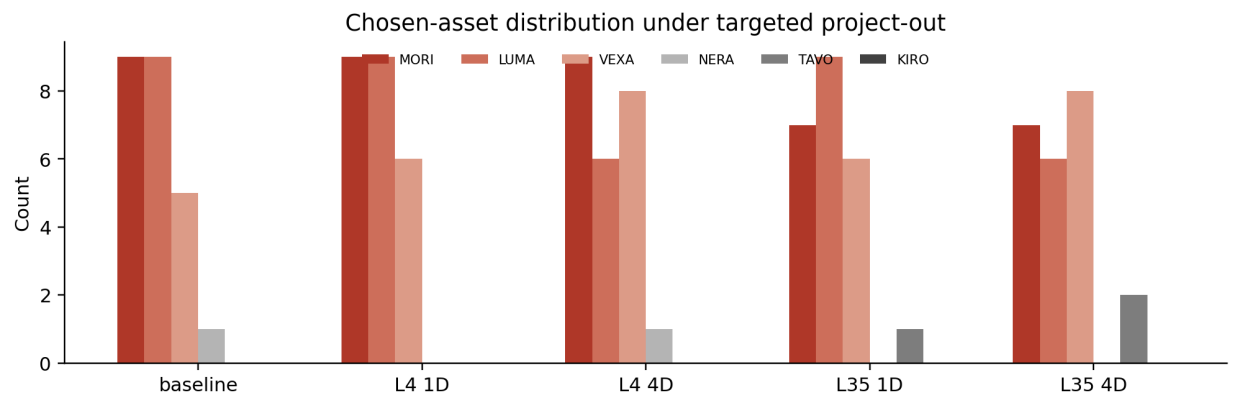
Behavioral Comparison



Each experiment compares the targeted project_out against its matched orthogonal random control on the same deterministic 24-prompt cohort.

Intervention	Tool-name change	Asset change	Mean spend delta	Mean token-count delta
Leader 1D p/o	0	0.083	6.25	349.4
Leader 1D ctrl	0.042	0.292	5.435	467
Leader 4D p/o	0	0.25	8.333	432.6
Leader 4D ctrl	0	0.167	3.125	402.9
Dispersion 1D p/o	0.042	0.125	4.348	353
Dispersion 1D ctrl	0	0.125	5.208	493.6
Dispersion 4D p/o	0.042	0.333	4.348	437.3
Dispersion 4D ctrl	0	0.125	6.25	489.2

Asset Surface



Targeted project-out conditions only. The full 4D subspace patches move the chosen-asset distribution more than the named 1D patches.

The concrete pattern is:

- Leader 1D: mostly preserves the baseline MORI/LUMA mix

- Leader 4D: pushes meaningfully toward VEXA
- Dispersion 1D: introduces one record_observation and modest extra VEXA/TAVO
- Dispersion 4D: produces the strongest asset redistribution, including TAVO and KIRO

Interpretation

Three conclusions:

1. The named 1D axes are real representation directions, but weak causal handles on final policy.
2. The broader 4D subspaces are more causally relevant than the single named PCs.
3. The L35 subspace is the strongest signal: Dispersion 4D clearly beats its matched control on asset changes (8/24 vs 3/24) and produces the largest targeted behavioral effect in the batch.

This is consistent with Phase 17's finding that the market encoding is broad and multi-dimensional. The 1D patches underperform; the 4D patches get stronger. That is what you would expect if the market summary is distributed across a small subspace rather than concentrated into one special direction.

Next Steps

The clearest next experiment is a joint patch: L4 top-4 + L35 top-4 applied together on the same cohort.

That would test whether the early leader/prominence summary and the later dispersion/unevenness summary are jointly necessary for the downstream decision. If the current pattern holds, the joint patch should be stronger than either single-layer patch alone.

The same deterministic stratified cohort and matched orthogonal controls should be kept for comparability.